

Work

Recognizing System Issues

1. Which problems have been spotted and what new problems arise due to those inefficiencies?

- There was low system usability because of slow automation processes that were challenging to manage.
- Frequent baggage backlogs and misplaced luggage because of unreliable switches within the conveyor belt systems.
- Delays and lost baggage resulting from excessive malfunctioning and maintenance expenses.
- Low level of coherence with the remaining baggage systems within the airlines.

2. Are there any vague or unfulfilled expectations?

- Absence of stakeholder engagement - the airlines did not actively participate in the system design at a strategic level.
- Overly optimistic expectations resulted in insufficient implementation testing.
- Insufficient breakdown resolution measures, such as manual backup procedures, meant that there was inadequate redundancy coverage.
- Lacking training materials and guides for computer usage for the airport personnel.

3. Gather Requirements: How might an analogous system operate at a different airport?

o A lot of successful airports, like as Changi (Singapore) and Schiphol (Amsterdam), use hybrid systems that combine manual and automated handling.

Prior to complete deployment, these airports conduct thorough testing processes.

In order to meet operational needs, they make sure that all relevant parties are involved, including airlines, maintenance crews, and airport employees.

4. Examine the video's workflows: Does the system successfully satisfy functional requirements?

o No, the system had trouble fulfilling its functional requirements.

o Mechanical problems frequently prevented bags from arriving reaching their destinations on schedule.

o The inefficient operation of the autonomous carts resulted in traffic jams and misdirected luggage.

o Rather than expediting luggage processing, the system increased delays.

5. To identify potential design flaws, use observation and document analysis.

- o There was no redundancy in the system; large-scale disruptions resulted from failures in one component.
- Inaccurate luggage tracking and routing were caused by poor software design.
- o There was no effective means to manually override the system or reroute conveyor belts and trolleys, and they were prone to mechanical breakdowns.

Redesign and Proposal for a Solution

1. Three Significant Enhancements to the Current System:

- o Hybrid Approach: To increase flexibility, combine manual handling and automation.
- o Modular Design: To avoid system-wide failures, implement independent sections that can operate independently.
- o Increased Stakeholder Involvement: During design and testing, collaborate closely with airlines, maintenance crews, and airport employees.

2. Use a Case Diagram or User Story to Explain Suggested Changes: To demonstrate how a hybrid system might operate, a diagram can be made with programs like draw.io or boards.com.

- o As an illustration, consider a procedure where bags with the highest priority are handled automatically and others are sorted by hand as needed.

3. Take into Account Agile/Iterative Approaches in Solutions: Execute in stages to provide testing and incremental enhancements.

Utilize input from passengers and airline employees to improve the system.